Standard Operating Procedure for Resolving BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Arista Access Switch WFD-ACC-31 at Whitefield POP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5907

Description

This Standard Operating Procedure (SOP) addresses the Root Cause Analysis (RCA) and resolution of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Arista access switch WFD-ACC-31, located at the Whitefield POP. The primary protocol in question is BGP, specifically the eBGP session established with the core router. The OSPF adjacency loss is likely due to a misconfiguration or hardware failure, while the ISIS LSP flooding is associated with high-scale network instability. The potential underlying causes include, but are not limited to, misconfigured routing policies, hardware issues, or software defects. The switch's access role, coupled with its location in the network topology, necessitates a comprehensive troubleshooting approach to isolate the root cause of the failure domain. The following SOP provides a structured process for identifying and resolving these issues on the Arista access switch.

Pre-requisite Checks

- Check: Verify that the Whitefield POP is accessible and the switch is reachable via SSH or console.

- Check: Verify that redundant links are functioning properly and that there are no hardware issues affecting the switch.

- Check: Verify that the eBGP session with the core router is established and stable.

Potential Root Causes

Cause	Probability
Misconfigured routing policies	High
Hardware issues	Medium
Software defects	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays an overview of the BGP session, including the number of peers, routes, and any errors encountered.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command provides details about the OSPF neighbor relationships, including the state of the adjacency and any
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errors encountered.

Command: show isis database, Expected Output: ISIS database
information, Description: This command displays the ISIS database,
including the LSPs and any errors encountered.
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Resolution Steps

Step 1: Step: Verify and correct misconfigured routing policies, Command: show run | inc policy, Description: This command displays the routing policies configured on the switch. Review the output to ensure that the policies are correctly configured and do not contribute to the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding.

Step 2: Step: Verify and correct hardware issues, Command: show inventory, Description: This command displays the switch's inventory, including the hardware components and their status. Review the output to identify any hardware issues that may be contributing to the failure domain.

Step 3: Step: Verify and correct software defects, Command: show version, Description: This command displays the switch's software version and any patches applied. Review the output to ensure that the switch is running the latest software version and that any known defects have been addressed.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information
with no errors, Description: This command verifies that the BGP
session is established and stable.

Command: show ospf neighbor, Expected Output: OSPF neighbor
information with no errors, Description: This command verifies that
the OSPF adjacency is established and stable.

Command: show isis database, Expected Output: ISIS database
information with no errors, Description: This command verifies that
the ISIS LSP flooding has been resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy startup-config running-config, Description: This command restores the original switch configuration, which may have been modified during the troubleshooting process.

- Step: Reset the BGP session, Command: clear bgp *, Description: This command resets the BGP session to its original state.